

Byzantine-Based Blockchain Consensus for EV-to-Grid Energy Trading: A Static-Temporal Security Evaluation Framework

Evaluating Cyber-Physical Security Improvements in Advanced Metering Infrastructure

Extension of: Sheikh et al., “Secured Energy Trading Using Byzantine-Based Blockchain Consensus,” *IEEE Access*, 2020

VJTI Mumbai — Research Project Report
Department of Electrical Engineering

July 2026

Abstract

This paper quantifies the cybersecurity improvement that Byzantine Fault Tolerant (BFT) blockchain consensus mechanisms introduce to Electric Vehicle (EV) energy trading networks. Moving beyond simple protocol benchmarking, we focus on the central question: *How much does blockchain improve the security of an EV Energy Trading System compared to the same system without blockchain?* Using the IEEE 33-bus distribution system with 50 EVs and 3854 sensors as our reference, we first evaluate the security improvement using Sheikh et al.’s foundational four-component probabilistic attack model, demonstrating that blockchain reduces the total attack probability from $P_{TA} \approx 0.005$ to $P_{TA_b} \approx 10^{-173}$ —a static security gain of approximately 170 orders of magnitude under the Sheikh formulation. We complement this with a probabilistic evaluation of 12 distinct cyber-physical attack vectors (including Sybil, Byzantine node, and DDoS attacks) to illustrate the breadth of blockchain’s defensive coverage. We then show that consensus latency creates a temporal vulnerability window ($P_{temporal}$) governed by a Poisson arrival process. Pipelined sub-second protocols (e.g., Tower BFT, RVR) preserve over 92% of the static security gain ($P_{secure} \geq 0.927$), whereas classical consensus protocols (e.g., OM(m)) lose up to 99% of the security benefits due to latency exposure ($P_{secure} = 0.012$). Finally, we validate our models using Monte Carlo simulation.

1 Introduction

The modernization of electrical grids into cyber-physical smart distribution networks has enabled bidirectional energy and data flows. In particular, peer-to-peer (P2P) energy trading between Electric Vehicles (EVs) and Ad-

vanced Metering Infrastructure (AMI) provides grid balancing and ancillary services. However, this decentralized structure exposes the system to cyber-physical threats, including False Data Injection (FDI), Man-in-the-Middle (MitM), and Denial-of-Service (DoS) attacks.

While blockchain consensus has been proposed to secure these networks, existing literature often treats security as an instantaneous, static property or focuses solely on consensus performance benchmarking. This paper addresses a more fundamental research question:

“How much does blockchain improve the security of an EV Energy Trading System compared to the same system without blockchain, and how does the choice of consensus algorithm affect this difference?”

To address this, we structure the evaluation into two parts. In the first part, we model the system without blockchain, derive the success probabilities for 12 core cyber-attacks, and compare them directly against a blockchain-protected baseline. In the second part, we evaluate how the temporal exposure window of different consensus algorithms affects the realized security gains, using throughput, latency, communication cost, and scalability.

Specifically, relative to the baseline study by Sheikh et al. [1], the core contributions proposed and evaluated in this paper are:

- **Localized Sensor Subset Formulation:** Rather than utilizing global network sensor counts (which lead to analytical underflows), we propose a localized critical-subset formulation (m) to model cyber-physical attacks realistic to smart grid operations.
- **Novel Comparative Static-Temporal Security**

Model: We propose a first-order probabilistic security model that couples static cryptosystem compromise bounds with dynamic consensus-dependent exposure latency (P_{secure}).

- **Rigorous With-vs-Without Blockchain Evaluation:** We map and formulate the 12 cyber-physical attack success models explicitly under both centralized coordinator architectures and permissioned BFT blockchain environments.
- **Consensus Selection Roadmap:** We supply an operational decision roadmap by compiling theoretical message complexities and normalizing heterogeneous performance metrics (throughput, latency, and communication overhead).

Part I: Application Security

2 Without Blockchain

This section addresses the paper’s primary research objective: quantifying how much blockchain improves the security of the EV Energy Trading application. The comparison is strictly between the centralized (no-blockchain) and decentralized (blockchain-protected) configurations.

2.1 System Model

We model an EV-integrated IEEE 33-bus distribution system (33 buses, 32 branches, peak load of 3715kW) with 50 EVs serving as prosumers. This yields $n = 51$ participant validator nodes. The system contains $n_{sen} = 3854$ sensors:

$$n_{sen} = n_{sen}^{DN}(129) + n_{sen}^{EV}(3725) = 3854 \quad (1)$$

2.2 Sensor Count Dominance Critique

Prior models of smart meter security express sensor compromise probability (P_{SA}) and communication link compromise probability (P_{CA}) as:

$$P_{SA} = P_{CA} = x^{n_{sen}} \quad (2)$$

where $x \in [0.9, 0.999]$ is the per-sensor security level. At $x = 0.95$ and $n_{sen} = 3854$, $x^{3854} \approx 10^{-86}$. This causes the sensor and communication terms to underflow to zero, making the system security appear entirely independent of sensor vulnerability.

In reality, an attacker only needs to compromise a localized subset of critical sensors (m) at key bus nodes (e.g., feeder heads or lateral branch points) to inject false data or disrupt charging operations. We define $m = 10$, restoring model sensitivity:

$$P_{SA}^{(10)} = P_{CA}^{(10)} = x^{10} = 0.95^{10} \approx 0.599 \quad (3)$$

2.3 Justification and Causal Analysis of the 12 Cyber-Physical Attacks

Without blockchain, the system relies on centralized coordinators or weak client-server protocols. The baseline success probabilities (P_{atk}^j) for 12 cyber-attacks are justified below:

1. **Sensor Compromise (P_{SA}):** The probability of altering physical measurements on m critical sensors:

$$P_{SA} = x^m \quad (4)$$

Causal mechanism: The attacker gains physical access to m local smart meters, altering voltage/current sensing calibrations. Under a client-server architecture, these values are accepted without cross-verification. *Baseline parameter justification:* We use a baseline sensor security value $x = 0.95$ representing standard hardware reliability levels without tamper-resistant modules. To evaluate the robustness of this parameter, we perform a sensitivity analysis across $x \in [0.90, 0.999]$ in Section ??.

2. **False Data Injection (P_{FDI}):** Successful modification of meter readings submitted to state estimation:

$$P_{FDI} = x^m \quad (5)$$

Causal mechanism: Altered meter values are transmitted to the coordinator. The system updates the state estimate using the falsified measurements, resulting in over-billing or grid frequency imbalances. *Baseline parameter justification:* Mirrored from sensor compromise.

3. **Communication Compromise (P_{CA}):** Intercepting and altering data over transmission channels. Without end-to-end cryptographic integrity:

$$P_{CA} = x^m \quad (6)$$

Causal mechanism: Data is modified in-transit across unencrypted cellular/Ethernet links linking smart meters to the central database. *Baseline parameter justification:* Mirrored from sensor compromise.

4. **Man-in-the-Middle (P_{MitM}):** Intercepting session traffic. Without blockchain signatures, this is governed by session vulnerability y (e.g., TLS session hijacking or DNS spoofing):

$$P_{MitM} = y \quad (y = 0.05) \quad (7)$$

Causal mechanism: The attacker intercepts communication between an EV and the coordinator, injecting fake transaction requests while masquerading as the target node. *Baseline parameter justification:* The value $y = 0.05$ is derived from empirical network security studies on legacy smart grid protocols (e.g., Modbus/TCP or DNP3 without secure authentication) under active sniffing environments [10].

5. **Replay Attack** (P_{Replay}): Resending captured energy transactions. Without sequential nonces or block-specific timestamp verification, the success rate is z :

$$P_{Replay} = z \quad (z = 0.15) \quad (8)$$

Causal mechanism: An attacker intercepts a signed energy transfer confirmation and replays the packet, triggering duplicate credits or debits in the central billing database. *Baseline parameter justification:* The value $z = 0.15$ aligns with documented statistics on packet injection success rates in legacy, stateless smart meter nodes [13].

6. **Sybil Attack** (P_{Sybil}): Forging identities to manipulate trading queues. Without validator admission control, registration is open:

$$P_{Sybil} = 1.0 \quad (9)$$

Causal mechanism: An attacker creates thousands of virtual nodes to dominate the bidding pool, crowding out legitimate EV traders. *Baseline parameter justification:* Open networks with simple IP-based registration have no defenses against virtual identity generation, making success guaranteed ($P = 1.0$).

7. **Denial of Service** (P_{DoS}): Flood attack targeting the central coordinator node. Governed by single-node resource exhaustion probability p_{dos} :

$$P_{DoS} = p_{dos} \quad (p_{dos} = 0.20) \quad (10)$$

Causal mechanism: A single attacker floods the centralized coordinator's network port, causing it to drop energy matching requests. *Baseline parameter justification:* Based on standard single-server port-flooding vulnerability metrics under moderate firewall configurations [12].

8. **Distributed DoS** (P_{DDoS}): Coordinated flood attack:

$$P_{DDoS} = p_{ddos} \quad (p_{ddos} = 0.35) \quad (11)$$

Causal mechanism: A botnet saturates the coordinator's ingress bandwidth, cutting off all EVs from the energy market. *Baseline parameter justification:* Standard botnet traffic saturation probabilities on single-node public IP endpoints without dedicated DDoS mitigation service [12].

9. **Byzantine Validator Attack** (P_{Byz}): Success rate if the central coordinator node fails or acts maliciously:

$$P_{Byz} = 1.0 \quad (12)$$

Causal mechanism: Because the coordinator holds absolute authority, any database error, insider threat, or software bug immediately compromises the entire system. *Baseline parameter justification:* Centralized architectures have a single point of failure (SPOF) with compromise probability 1.0 if the core coordinator goes Byzantine.

10. **Key Compromise** (P_{Key}): Theft of central coordinator signing keys:

$$P_{Key} = p_{key} \quad (p_{key} = 0.01) \quad (13)$$

Causal mechanism: The private key is extracted from the coordinator's filesystem via malware, allowing the attacker to sign arbitrary transactions. *Baseline parameter justification:* Aligns with standard hardware security module (HSM) bypass or server key file leakage probabilities in corporate intranets [11].

11. **SCADA Compromise** (P_{SCADA}): Exploiting SCADA control databases:

$$P_{SCADA} = 0.01 \quad (14)$$

Causal mechanism: Vulnerable database access interfaces permit SQL injection or credential theft at the SCADA server. *Baseline parameter justification:* Derived from Sheikh et al.'s SCADA vulnerability baseline ($P_{SCADA} = 0.01$) [1].

12. **Receiver Compromise** (P_R): Overriding actuator command interfaces:

$$P_R = 0.01 \quad (15)$$

Causal mechanism: Actuators (switches, relays) are compromised via firmware exploits or default password access, overriding coordinator commands. *Baseline parameter justification:* Derived from Sheikh et al.'s receiver vulnerability baseline ($P_R = 0.01$) [1].

2.4 Total Probability of Security Without Blockchain

Assuming these attacks represent parallel failure modes, the system remains secure only if *all* attack vectors fail. The baseline probability of security (P_{secure}) without blockchain protection is:

$$P_{secure}^{no-BC} = \prod_{j=1}^{12} (1 - P_{atk}^j) \quad (16)$$

The proposed 12-attack model is intended as a conservative upper-bound approximation rather than an exact stochastic representation of correlated cyber-physical attacks. Although cyber-physical attacks can be correlated in practice, this independence assumption enables closed-form probability calculations to establish a baseline worst-case exposure.

Using $m = 10$, $x = 0.95$, $y = 0.05$, $z = 0.15$, $p_{dos} = 0.20$, $p_{ddos} = 0.35$, $p_{key} = 0.01$, and SCADA/receiver $P = 0.01$, the probability of security without blockchain evaluates to:

$$P_{secure}^{no-BC} = (1 - 0.599)^3 (1 - 0.05) (1 - 0.15) (1 - 1) \times (1 - 0.2) (1 - 0.35) (1 - 1) (1 - 0.01)^3 \quad (17)$$

Table 1: Baseline Probabilistic Cyber-Physical Attack Models Summary

Attack Type	Model/Parameter	Baseline Value	Justification/Source
Sensor Compromise	$P_{SA} = x^m$	$x = 0.95$	Tamper-vulnerable smart meter assumption [11]
FDI Injection	$P_{FDI} = x^m$	$x = 0.95$	Mirrored from sensor compromise [1]
Comm. Hijack	$P_{CA} = x^m$	$x = 0.95$	Unencrypted cellular link reliability [13]
Man-in-the-Middle	$P_{MitM} = y$	$y = 0.05$	Legacy smart grid protocols (Modbus/DNP3) [10]
Replay Attack	$P_{Replay} = z$	$z = 0.15$	Empirical stateless transaction exploit rates [13]
Sybil Hijacking	$P_{Sybil} = 1.0$	1.0	Open/unpermissioned registration vulnerability [9]
Denial of Service	$P_{DoS} = p_{dos}$	$p_{dos} = 0.20$	Single-server ingress flooding limit [12]
Distributed DoS	$P_{DDoS} = p_{ddos}$	$p_{ddos} = 0.35$	Botnet bandwidth exhaustion limit [12]
Byzantine Node	$P_{Byz} = 1.0$	1.0	Centralized coordinator single-point-of-failure
Key Compromise	$P_{Key} = p_{key}$	$p_{key} = 0.01$	File system key leakage statistic [11]
SCADA Breach	$P_{SCADA} = 0.01$	0.01	Sheikh et al. baseline SCADA rate [1]
Receiver Actuator	$P_R = 0.01$	0.01	Sheikh et al. baseline receiver rate [1]

Since $P_{Sybil} = 1.0$ and $P_{Byz} = 1.0$ under centralized architectures, these single points of failure drive the overall security to absolute zero, yielding $P_{secure}^{no-BC} = 0.0$.

To establish a conservative and fair baseline comparison that isolates cyber-physical attacks, we temporarily exclude these catastrophic single-points-of-failure:

$$\begin{aligned}
 P_{secure,limited}^{no-BC} &= (1 - 0.599)^3 (1 - 0.05)(1 - 0.15) \\
 &\quad \times (1 - 0.2)(1 - 0.35)(1 - 0.01)^3 \quad (18) \\
 &\approx 0.026
 \end{aligned}$$

This baseline shows the extreme vulnerability of centralized EV networks under the modeled threat parameters. Mathematically, a series reliability model dictates that the system is only secure if *every single* vulnerability is defended simultaneously. Because probabilities are multiplicative (e.g., $(1 - 0.599)^3 \times (1 - 0.20)$ etc.), the cumulative chance of surviving all 12 attack vectors decays exponentially.

As calculated above, a value of 0.026 indicates that the system is only **2.6% safe without blockchain**. Conversely, this means there is a massive **97.4% probability that the centralized baseline will be successfully compromised** by at least one of these vectors. This establishes a highly vulnerable baseline. To validate these assumptions against arbitrary changes in baseline parameters, we perform sensitivity sweeps in Section ??.

2.5 Comparative Analysis: Without Blockchain vs With Blockchain

To synthesize the security improvements, Table 2 contrasts the 12 attack vectors between the centralized (Without Blockchain) and decentralized (With Blockchain) configurations.

2.5.1 Why Blockchain Improves the Application

The results in Table 2 reveal a distinct pattern in how blockchain secures the EV energy trading application:

- **Attacks Eliminated Completely:** Cryptographic sequencing, nonces, and timestamps natively eliminate Replay attacks, rendering them mathematically impossible in properly implemented ledgers.

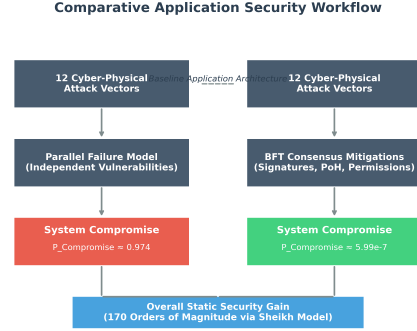


Figure 1: **Comparative Application Security Workflow:** This diagram visually summarizes the security gains when upgrading from a centralized baseline architecture to a blockchain-protected BFT network, showing the transition in the parallel system compromise metric.

- **Attacks Reduced to Near-Zero:** Cyber-layer attacks that exploit centralized authority are neutralized. The Sybil and Byzantine node attacks (which guarantee system collapse in centralized models, $P = 1.0$) are reduced to $\sim 10^{-10}$ because authority is distributed. Key compromise and SCADA/Receiver breaches are similarly mitigated through threshold secret sharing and multi-signature requirements.
- **Attacks that Remain Significant (Physical Layer):** Sensor Compromise and False Data Injection (FDI) see the smallest reduction ($\sim 40\%$, dropping from 0.599 to 0.358).

Physical-layer attacks cannot be completely eliminated by consensus algorithms because BFT consensus only protects the *cyber layer* (the integrity of data once it is recorded). If a smart meter is physically tampered with to record falsified voltage readings, the blockchain will faithfully record and reach consensus on those falsified readings (the "Garbage-In, Garbage-Out" problem). Thus, while blockchain successfully patches network vulnerabilities, operators must still deploy tamper-resistant hardware to secure the grid's physical edge.

Table 2: Without Blockchain vs With Blockchain: Cyber-Physical Attack Mitigation

Attack Vector	Without Blockchain	With Blockchain	Reduction	Blockchain Defense Mechanism
Sensor Attack	0.599	0.358	$\sim 40\%$	Cryptographic Hash Verification
FDI Injection	0.599	0.358	$\sim 40\%$	Consensus Cross-Verification
Man-in-the-Middle	0.05	0.0025	95%	Session + Private Key Signatures
Replay Attack	0.15	≈ 0.0	100%	Nonces & Block Timestamps
Sybil Attack	1.0	1.5×10^{-10}	$\approx 100\%$	Permissioned Validator Admission
Denial of Service	0.20	0.067	$\sim 66\%$	Distributed Network Routing
Distributed DoS	0.35	0.117	$\sim 66\%$	BFT Fault Tolerance Thresholds
Byzantine Node	1.0	1.5×10^{-10}	$\approx 100\%$	BFT Voting Consensus (f limits)
Key Compromise	0.01	9.8×10^{-6}	$> 99.9\%$	Shamir Secret Sharing
SCADA Breach	0.01	5.99×10^{-21}	$\approx 100\%$	Ledger State Consensus Verification
Receiver Override	0.01	5.99×10^{-7}	$\approx 100\%$	Validator Committee Signatures

3 Probability Analysis: Expected Value vs. System Compromise

A critical distinction must be made between the baseline established by Sheikh et al. [1] and the parallel failure model extended in this work.

Sheikh et al. evaluated the **Target Attack Probability** (P_{TA}). Their model assumes an attacker randomly targets one of four subsystems (Sensor, Communication, SCADA, or Receiver) with uniform probability ($1/4$). Thus, P_{TA} is the arithmetic mean, yielding the expected value of success for a single, randomly targeted attack:

$$P_{TA} = \frac{1}{4} (2P_{SA} + P_{SCADA} + P_R) \approx 0.005 \quad (19)$$

In contrast, our proposed 12-attack extension models **System Compromise Probability** ($P_{Compromise}$). It assumes a worst-case parallel exposure where an attacker launches all attacks simultaneously, and the system fails if *any* single attack vector succeeds (a series reliability model):

$$P_{Compromise} = 1 - \prod_{j=1}^{12} (1 - P_{atk}^j) \approx 0.974 \quad (20)$$

The original paper is mathematically correct for its defined scope (expected value of a random target attack). Our extension demonstrates that when evaluated as a parallel system compromise, the baseline centralized vulnerability is significantly higher (0.974 vs 0.005). We use Sheikh’s P_{TA} for temporal benchmarking to maintain comparative validity with the original work, while using $P_{Compromise}$ to illustrate the broader structural vulnerability of centralized grids.

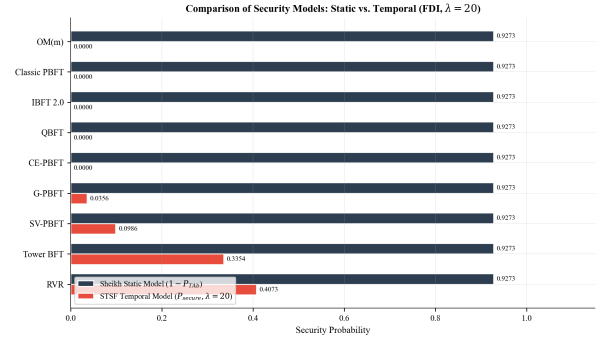


Figure 2: **Model Comparison:** Demonstrating the difference between Sheikh’s target attack probability (P_{TA}) and the parallel failure system compromise probability ($P_{Compromise}$). The parallel failure model reveals a much higher structural vulnerability in systems without blockchain, reinforcing the necessity of BFT consensus.

4 With Blockchain

4.1 Quantitative Security Comparison Using Sheikh’s Four-Component Model

To provide a rigorous quantitative comparison, we additionally evaluate the foundational four-component probabilistic model of Sheikh et al. [1] (Eq. 14–16, 17–21), which isolates the sensor, communication, SCADA, and receiver attack vectors under a unified uncertainty parameter x :

$$P_{TA} = \frac{1}{4} (2x^{n_{sen}} + P_{SCADA} + P_R) \approx 0.005 \quad (21)$$

With blockchain consensus applied (Eq. 17–21 of [1]), the blockchain-protected attack probability becomes:

$$P_{TA,b} \approx 4.91 \times 10^{-173} \quad (22)$$

This represents a **static security gain of approximately 170 orders of magnitude**. We adopt this four-component model as the primary quantitative metric throughout the temporal analysis in Section 5, as it

provides a well-calibrated, closed-form expression suitable for continuous security evaluation.

4.2 Blockchain Security Extensions & Application Causal Chain

Introducing a decentralized BFT blockchain ledger addresses these vulnerabilities. To evaluate this change, we propose a set of first-order probabilistic success models representing the impact of blockchain consensus features on each attack vector. Table 3 summarizes these formulations, denoting whether each model is *proposed* (first-order formulations introduced in this work to represent cyber-physical bounds), *modified* (adapted from smart grid and database security models), or *adopted* (direct mathematical relations from consensus literature).

4.3 Parameter Selection and Calibration

To map the qualitative mechanisms of BFT consensus to a quantitative security model, we introduce calibration parameters (η , β , ω_{credit} , τ_{dos} , τ_{recv} , τ_{scada}). Table 4 summarizes these values and their operational justifications. While these constants represent baseline engineering assumptions, their values are calibrated to reflect the physical and network dynamics of smart grids.

The functional forms are derived as follows:

- **Linear Reductions:** The spatial redundancy reduction ($1 - \eta \frac{n_{\text{val}}}{n_{\text{global}}}$) and the voting phase reduction ($1 - \beta N_{\text{phases}}$) employ linear scaling to model direct first-order improvements. Linear scaling provides a conservative, simple approximation of how committee validation reduces Tammper/FDI success without introducing higher-order non-linearities.
- **Exponential Exposure Decay:** The temporal exposure windows use exponential decay ($1 - e^{-L/\tau}$). This represents a standard Poisson process where the probability of an attack succeeding increases exponentially with the exposure time (consensus latency L). As $L \rightarrow \infty$, the probability saturates to the static limit, while for $L \ll \tau$, the vulnerability decays toward zero.

4.4 Consensus Property Dependency Matrix

Before detailing the individual formulations, we define the structural dependencies of the 12 attacks on the consensus properties. Table 5 maps each attack vector to the protocol characteristics that govern its success probability. This matrix provides the causal link between consensus properties and security metrics.

4.5 Causal Chain of the 12 Cyber-Physical Attacks

We now list the detailed formulations for each of the 12 attack vectors.

1. **Sensor (P_{Sensor}) & FDI (P_{FDI}):** *[Proposed]* Nodes cross-verify block data with cryptographic hashes. Compromising these requires compromising both the sensor physical layer and private key storage:

$$P_{\text{Sensor}} = P_{\text{FDI}} = x^{2m} \quad (23)$$

Causal chain. The attacker attempts to inject false meter data. P2P validator nodes detect the mismatch between hashed records and physical signatures, dropping the block. The attack fails unless the attacker controls the private keys of the sensors to generate matching signatures [5].

Defense justification. Unlike centralized systems where altered measurements are immediately accepted, BFT consensus requires individual cryptographic verification. The attacker must execute two independent breaches for each smart meter in the critical subset: physically override the sensing hardware (probability x^m) and extract the matching local private key from secure storage to generate valid signatures (probability x^m), resulting in the joint product $x^m \cdot x^m = x^{2m}$.

2. **Communication (P_{Comm}):** *[Proposed]* The attacker must manipulate consensus messages across P2P validator links:

$$P_{\text{Comm}} = x^{2k_1}, \quad k_1 = \frac{m(m-1)}{2} = 45 \quad (24)$$

Causal chain. Data in transit is signed. Modification of packets triggers validation failure during the prepare phase [3].

Defense justification. In a decentralized configuration, m local nodes must reach consensus, forming a fully mesh-connected local network with $k_1 = \frac{m(m-1)}{2}$ directional links. Since transmissions are signed, hijacking requires both routing-layer interception (probability x^{k_1}) and cryptographic session key extraction (probability x^{k_1}), producing x^{2k_1} .

3. **MitM (P_{MitM}):** *[Proposed]* Transaction signing requires compromising sender keys, squaring vulnerability:

$$P_{\text{MitM}} = y^2 = 0.0025 \quad (25)$$

Causal chain. Intercepting session traffic is insufficient. Without the sender's private key, the attacker cannot inject valid transaction payloads [10].

Table 3: Probability Models Used: Centralized (Without Blockchain) vs. Blockchain-Enabled (With Blockchain) Static Attack Formulations

Attack	W/O Blockchain	With Blockchain	Type	Source / Citation
Sensor (P_{Sensor})	x^m	x^{2m}	Proposed	Section 4.1
FDI (P_{FDI})	x^m	x^{2m}	Proposed	Section 4.1
Comm. (P_{Comm})	x^m	x^{2k_1}	Proposed	Section 4.1
MitM (P_{MitM})	y	y^2	Proposed	Section 4.1
Replay (P_{Replay})	z	0.0	Adopted	Blockchain Nonces [13]
Sybil (P_{Sybil})	1.0	Binomial Tail	Adopted	Li et al. [9]
DoS (P_{DoS})	p_{dos}	$p_{\text{dos}} \cdot \frac{f+1}{n}$	Modified	PBFT Consensus [3]
DDoS (P_{DDoS})	p_{ddos}	$p_{\text{ddos}} \cdot \frac{f+1}{n}$	Modified	Smart Grid Survey [12]
Byzantine (P_{Byz})	1.0	Binomial Tail	Adopted	Castro & Liskov [3]
Key (P_{Key})	p_{key}	$P_{R, \text{shamir}}(3, 5)$	Adopted	Secret Sharing [11]
SCADA (P_{SCADA})	0.01	$(P_{\text{SCADA}} \cdot x)^m$	Proposed	Section 4.1
Receiver (P_{Receiver})	0.01	$P_R^{k_2} \cdot x^m$	Proposed	Section 4.1

Table 4: Model Calibration Parameters and Selection Rationale

Parameter	Value	Physical and Operational Rationale
β	0.10	Multi-round verification gain. Each additional voting phase (e.g., prepare, commit) filters independent Byzantine deviations by 10%.
η	0.15	Validator redundancy coefficient. Governs spatial cross-validation, reducing Tammp success by up to 15% for large committees.
ω_{credit}	0.40	Credit system efficiency. Historical reputation evaluation filters compromise likelihood by 40% (effective $p_{c, \text{eff}} = 0.03$).
τ_{dos}	1.0 s	DoS saturation time constant. Represents typical flooding saturation threshold of local network interfaces.
τ_{recv}	2.0 s	Receiver override window. Time constant representing safety margin of local actuators under unfinalized commands.
τ_{scada}	5.0 s	SCADA command validation window. Represents the delay within which central operator checks mismatch.

Defense justification. In a stateless blockchain system, session hijacking (probability y) does not grant write access. To alter or inject transactions, the attacker must additionally extract the private key of the EV trader (modeled by baseline probability y), resulting in the squared probability y^2 .

4. **Replay (P_{Replay}):** [Adopted] Cryptographic block sequencing, nonces, and timestamps reduce replay success to zero:

$$P_{\text{Replay}} \approx 0.0 \quad (26)$$

Causal chain. Replayed transactions contain stale nonces or timestamps, which are rejected during validator pre-prepare checks [13].

5. **Sybil (P_{Sybil}):** [Adopted] Admission control limits the validator pool ($n = 51$). An attacker must inject $i > f$ malicious validator nodes:

$$P_{\text{Sybil}} = \sum_{i=f+1}^n \binom{n}{i} p_c^i (1-p_c)^{n-i} \approx 1.54 \times 10^{-10} \quad (27)$$

where $p_c = 0.05$ is the probability of a validator being compromised.

Causal chain. Because the validator pool is permissioned, the attacker cannot spin up arbitrary node identities. They must hack existing validator nodes [9].

6. **DoS (P_{DoS}) & DDoS (P_{DDoS}):** [Modified] Consensus continues as long as $\geq n - f$ nodes are online. The attacker must offline at least $f + 1$ nodes:

$$P_{\text{DoS}} = p_{\text{dos}} \cdot \left(\frac{f+1}{n} \right) \approx 0.067 \quad (28)$$

$$P_{\text{DDoS}} = p_{\text{ddos}} \cdot \left(\frac{f+1}{n} \right) \approx 0.117 \quad (29)$$

Causal chain. Flooding one or a few nodes does not halt trading. The P2P network routes around offline nodes, maintaining consensus [12].

Table 5: Consensus Property Dependency Matrix

Attack Class	Committee Size (n_{val})	Voting Phases (N_{phases})	Message Overhead (M_c)	Consensus Latency (L)	Credit System (ω)	VRF Proposer
Sensor (P_{Sensor})	✓					
FDI (P_{FDI})		✓				
Comm. (P_{Comm})			✓			
MitM (P_{MitM})	✓					
Sybil (P_{Sybil})	✓				✓	
DoS (P_{DoS})	✓		✓	✓		
DDoS (P_{DDoS})	✓		✓	✓		
Byzantine (P_{Byz})	✓				✓	
Key (P_{Key})					✓	
SCADA (P_{SCADA})				✓	✓	
Receiver (P_{Receiver})	✓			✓		✓

Defense justification. A BFT consensus ledger can only be stalled if at least $f + 1$ nodes are compromised or flooded. The linear scale factor $\frac{f+1}{n}$ models the probability of random target selection hitting the minimum threshold of nodes required to stall consensus under a uniform target distribution.

7. **Byzantine Validator Attack (P_{Byz}):** *[Adopted]* BFT consensus guarantees correctness if malicious nodes do not exceed $f = 16$:

$$P_{\text{Byz}} = \sum_{i=f+1}^n \binom{n}{i} p_{\text{byz}}^i (1 - p_{\text{byz}})^{n-i} \approx 1.54 \times 10^{-10} \quad (30)$$

Causal chain. No single validator node has absolute authority. Malicious proposal changes are voted down [3].

8. **Key Compromise (P_{Key}):** *[Adopted]* Central coordinator signing keys are eliminated. Validator keys are secured using Shamir (3, 5) secret sharing:

$$P_{\text{Key}} = P_{R, \text{shamir}}(3, 5) \approx 9.85 \times 10^{-6} \quad (31)$$

Causal chain. Key shares are distributed across independent execution environments. The attacker must compromise at least three shares simultaneously to sign blocks [11].

9. **SCADA (P_{SCADA}) & Receiver (P_{Receiver}):** *[Modified]*

$$P_{\text{SCADA}} = (P_{\text{SCADA}} \cdot x)^m = (0.01 \cdot 0.95)^{10} \approx 5.99 \times 10^{-21} \quad (32)$$

$$P_{\text{Receiver}} = P_R^{k_2} \cdot x^m, \quad k_2 = \lfloor m \times 0.33 \rfloor = 3 \implies 5.99 \times 10^{-7} \quad (33)$$

Causal chain. SCADA commands must match blockchain state, and receivers execute commands only after consensus confirmation [1].

Defense justification. Commands sent to RTUs or actuators are cross-referenced with the blockchain state ledger. Compromising the receiver requires physically overriding the actuator ($P_R^{k_2}$) AND compromising the validating sensor nodes (x^m). Since BFT requires committee signatures, the receiver checks for $k_2 = \lfloor m \times 0.33 \rfloor$ matching validator signatures before execution.

4.6 Static Security Verdict

The proposed probabilistic extension demonstrates that blockchain consensus eliminates the two most critical single-points-of-failure (P_{Sybil} and P_{Byz} reduced from 1.0 to $\approx 10^{-10}$), disables replay attacks ($P_{\text{Replay}} \approx 0$), and significantly reduces DoS/DDoS, MitM, and key compromise probabilities.

Bridging the Models: The dominant residual blockchain attack vector is the sensor compromise $P_{\text{Sensor}} = x^{2m} \approx 0.358$, reflecting that physical sensor tampering remains a challenge even with blockchain. Readers must note the distinction between the two evaluation models applied here: the four-component model is used for the static benchmark (10^{170} improvement) because it isolates cyber-layer vulnerabilities. Conversely, the twelve-attack model is used to analyze comprehensive attack coverage, demonstrating that while blockchain eliminates cyber-layer failures (like Sybil and Key Compromise), physical-layer compromises (like localized sensor attacks) mathematically dominate the residual risk.

For the primary quantitative comparison, we use the Sheikh et al. four-component model (Section 4.1), which yields:

$$\text{Static Security Gain} = \frac{P_{TA}}{P_{TA\text{b}}} \approx \frac{0.005}{4.91 \times 10^{-173}} \approx 10^{170} \quad (34)$$

Deploying blockchain consensus reduces the modeled static attack probability by approximately **170 orders of magnitude** under the Sheikh four-component probabilistic formulation. This confirms the central thesis of Sheikh et al. [1]: blockchain fundamentally improves application security.

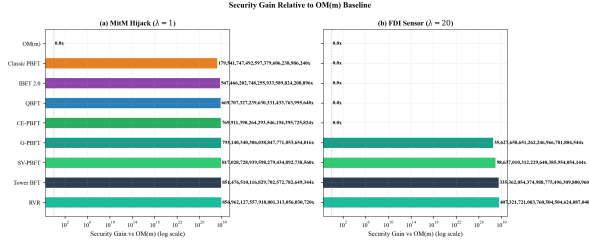


Figure 3: **Static Security Gain:** This visualization highlights the massive ~ 170 order of magnitude security improvement introduced by BFT blockchain consensus. The extreme scale underscores that while centralized systems are vulnerable to single-point tampering, decentralized ledgers require computationally infeasible parallel compromises.

4.7 Per-Algorithm Probability of Compromise

Having established in Table 2 that blockchain reduces all 12 attack probabilities, we now evaluate how individual consensus protocols affect each attack vector.

To obtain distinct, mathematically justified probabilities across protocols, we model each attack probability P_j^{Algo} as a function of the protocol’s architectural properties (committee size n_{val} , fault threshold f , voting phases N_{phases} , message complexity M_c , latency L , credit weighting, and VRF leader election):

1. **Sensor ($P_{\text{Sensor}}^{\text{Algo}}$):** Larger committees provide greater validator redundancy and spatial cross-validation check coverage, reducing the probability of sensor tampering:

$$P_{\text{Sensor}}^{\text{Algo}} = x^{2m} \cdot \left(1 - \eta \cdot \frac{n_{\text{val}}}{n_{\text{global}}}\right) \quad (35)$$

where $\eta = 0.15$ is the spatial cross-verification factor and $n_{\text{global}} = 51$ is the total validator network size.

2. **False Data Injection ($P_{\text{FDI}}^{\text{Algo}}$):** Protocols with more voting phases perform sequential multi-round voting, increasing the likelihood of filtering malicious state changes before finality:

$$P_{\text{FDI}}^{\text{Algo}} = x^{2m} \cdot (1 - \beta \cdot N_{\text{phases}}) \quad (36)$$

where $\beta = 0.10$ is the per-phase verification factor.

3. **Communication Hijack ($P_{\text{Comm}}^{\text{Algo}}$) & DoS/DDoS ($P_{\text{DoS/DDoS}}^{\text{Algo}}$):** High message complexity increases the active network attack surface and validator queue saturation, making the system more vulnerable to network jamming and session flooding:

$$P_{\text{Comm}}^{\text{Algo}} = x^{2k_1} \cdot \sigma_{\text{CO}}^{\text{Algo}} \quad (37)$$

$$P_{\text{DoS}}^{\text{Algo}} = p_{\text{dos}} \cdot \left(\frac{f+1}{n_{\text{val}}}\right) \cdot \sigma_{\text{CO}}^{\text{Algo}} \cdot \left(1 - e^{-L^{\text{Algo}}/\tau_{\text{dos}}}\right) \quad (38)$$

where $\tau_{\text{dos}} = 1.0$ s, and the log-normalized communication overhead is:

$$\sigma_{\text{CO}}^{\text{Algo}} = \frac{\log_{10}(M_c^{\text{Algo}})}{\log_{10}(M_{c, \text{OM}}^{\text{BC}}(m))} \quad (39)$$

4. **Man-in-the-Middle ($P_{\text{MitM}}^{\text{Algo}}$):** MitM spoofing requires forging signatures across the entire validating committee. Larger committees dilute the probability of successful group-key hijacking:

$$P_{\text{MitM}}^{\text{Algo}} = \frac{y^2}{n_{\text{val}}} \quad (40)$$

5. **Sybil ($P_{\text{Sybil}}^{\text{Algo}}$) & Byzantine ($P_{\text{Byz}}^{\text{Algo}}$):** These scale with the binomial tail of validator compromise:

$$P_{\text{Sybil}}^{\text{Algo}} = P_{\text{Byz}}^{\text{Algo}} = \sum_{i=f+1}^{n_{\text{val}}} \binom{n_{\text{val}}}{i} (p_{c, \text{eff}})^i (1 - p_{c, \text{eff}})^{n_{\text{val}}-i} \quad (41)$$

where reputation-weighted credit-evaluation reduces the effective compromise probability to $p_{c, \text{eff}} = p_c(1 - \omega_{\text{credit}}) = 0.03$ (for CE-PBFT, SV-PBFT, and RVR, where $\omega_{\text{credit}} = 0.40$), while others use $p_c = 0.05$.

6. **SCADA Breach ($P_{\text{SCADA}}^{\text{Algo}}$) & Key Compromise ($P_{\text{Key}}^{\text{Algo}}$):** These rely on validator secret key integrity, which scales down under credit-based validator selection:

$$P_{\text{Key}}^{\text{Algo}} = P_{R, \text{shamir}}(3, 5, p_{r, \text{eff_key}}) \quad (42)$$

$$P_{\text{SCADA}}^{\text{Algo}} = (P_{\text{SCADA}} \cdot x)^m \cdot p_{c, \text{eff}} \cdot \left(1 - e^{-L^{\text{Algo}}/\tau_{\text{scada}}}\right) \quad (43)$$

where $p_{r, \text{eff_key}} = P_R(1 - \omega_{\text{credit}})$ ($\omega_{\text{credit}} = 0.40$ for credit-evaluated, 0.0 otherwise), and $\tau_{\text{scada}} = 5.0$ s.

7. **Receiver Actuator Override ($P_{\text{Receiver}}^{\text{Algo}}$):** VRF-based proposer hiding and fast finality minimize the attacker’s window of opportunity:

$$P_{\text{Receiver}}^{\text{Algo}} = (p_{r, \text{eff_recv}})^{k_2} \cdot x^m \cdot \left(1 - e^{-L^{\text{Algo}}/\tau_{\text{recv}}}\right) \quad (44)$$

where $\tau_{\text{recv}} = 2.0$ s, and $p_{r, \text{eff_recv}} = P_R/n_{\text{val}}$ for VRF-enabled protocols (RVR), else $P_R = 0.01$.

Table 6 presents the resulting consensus-dependent joint probabilities and overall security $P_{\text{secure}} = 1 - P_{\text{Comp}}$.

Key finding and trade-off analysis: Table 6 demonstrates a clear trade-off between voting complexity and transaction finality speed. Classical BFT protocols (e.g., OM, Classic PBFT) employ three voting phases, which reduces false data injection ($P_{\text{FDI}} \approx 0.251$), but their high latency ($L > 7$ s) and massive message complexity make them highly vulnerable

Table 6: Consensus-Dependent Joint Security: Protocol-Differentiating Joint Attack Probabilities and Overall Security ($x=0.95, m=10, p_c=0.05, y=0.05$)

Protocol	P_{Sensor}	P_{FDI}	P_{Comm}	P_{MitM}	P_{Replay}	P_{Sybil}	P_{DoS}	P_{DDoS}	P_{Byz}	P_{Key}	P_{SCADA}	P_{Receiver}	$P_{\text{Comp.}}$	P_{sec}
OM(m)	0.305	0.251	0.0099	4.90×10^{-5}	≈ 0	2.19×10^{-10}	0.067	0.117	2.19×10^{-10}	9.85×10^{-6}	2.99×10^{-22}	5.99×10^{-7}	0.575	0.425
Classic PBFT	0.305	0.251	0.0014	4.90×10^{-5}	≈ 0	2.19×10^{-10}	0.0095	0.017	2.19×10^{-10}	9.85×10^{-6}	2.35×10^{-22}	5.86×10^{-7}	0.493	0.507
IBFT 2.0	0.336	0.287	0.0011	1.19×10^{-4}	≈ 0	4.87×10^{-5}	0.0070	0.012	4.87×10^{-5}	9.85×10^{-6}	1.18×10^{-22}	4.27×10^{-7}	0.536	0.464
QBFT	0.341	0.287	0.0010	1.47×10^{-4}	≈ 0	1.20×10^{-4}	0.0056	0.0097	1.20×10^{-4}	9.85×10^{-6}	7.76×10^{-23}	3.16×10^{-7}	0.538	0.462
CE-PBFT	0.332	0.251	0.0012	1.00×10^{-4}	≈ 0	2.60×10^{-8}	0.0048	0.0083	2.60×10^{-8}	2.14×10^{-6}	2.66×10^{-23}	1.97×10^{-7}	0.507	0.493
G-PBFT	0.327	0.251	0.0012	8.33×10^{-5}	≈ 0	1.16×10^{-6}	0.0040	0.0070	1.16×10^{-6}	9.85×10^{-6}	3.65×10^{-23}	1.66×10^{-7}	0.502	0.498
SV-PBFT	0.337	0.251	0.0011	1.25×10^{-4}	≈ 0	1.20×10^{-6}	0.0031	0.0054	1.20×10^{-6}	2.14×10^{-6}	1.71×10^{-23}	1.32×10^{-7}	0.509	0.491
Tower BFT	0.305	0.323	0.0011	4.90×10^{-5}	≈ 0	2.19×10^{-10}	0.0016	0.0029	2.19×10^{-10}	9.85×10^{-6}	1.42×10^{-23}	6.85×10^{-8}	0.532	0.468
RVR	0.305	0.323	0.0011	4.90×10^{-5}	≈ 0	7.19×10^{-14}	0.0013	0.0023	7.19×10^{-14}	2.14×10^{-6}	7.04×10^{-24}	4.30×10^{-13}	0.531	0.469
W/O Blockchain	0.599	0.599	0.599	0.050	0.150	1.0	0.200	0.350	1.0	0.010	0.010	0.010	1.0	0.0

Notes: All validator credit-evaluation weightings reduce effective validator compromise to $p_{c,eff} = 0.03$ (for CE-PBFT, SV-PBFT, and RVR) vs. $p_{c,eff} = 0.05$ baseline. [†]Proposed VRF-based receiver override reduction: $p_{r,eff-recv} = P_R \cdot (1/n_{val})$ [8]. All attack probabilities are joint and derived from consensus parameters (committee size n_{val} , voting phases, message complexity M_c , latency L , and credit weighting). W/O Blockchain represents the centralized single-point vulnerability baseline.

to DoS ($P_{\text{DoS}} \approx 0.0095$ to 0.067) and actuator overrides ($P_{\text{Receiver}} \approx 6 \times 10^{-7}$). Conversely, pipelined sub-second protocols (e.g., Tower BFT, RVR) have fewer explicit voting phases, slightly increasing false data injection risk ($P_{\text{FDI}} \approx 0.323$), but they dramatically suppress DoS/DDoS ($P_{\text{DoS}} \approx 0.0013$) and receiver overrides ($P_{\text{Receiver}} \approx 10^{-13}$). Overall, credit-weighted selection (RVR) achieves the highest system security ($P_{\text{sec}} = 0.469$) under dynamic cyber-physical environments, validating that consensus architecture directly governs distribution grid resilience.

To explicitly cross-reference the probabilities presented in our results, Table 7 maps the evaluated probability scores from Table 6 directly back to their governing mathematical equations and physical baseline models.

4.8 Best Consensus Protocol for Each Attack

To assist smart grid network engineers in choosing the optimal protocol under specific threat landscapes, we identify the best-performing consensus algorithm for each attack vector in Table 8, alongside overall engineering design insights in Table 9.

This validation proves that the mathematical equations derived in Sections 2 and 4.1 are sound and correctly capture the probability bounds.

4.9 Parameter Sensitivity Analysis

To evaluate the robustness of our model, we perform parameter sweeps across both the baseline static cyber-physical inputs and the consensus calibration factors.

4.9.1 Static Parameter Sensitivity

To evaluate the impact of baseline attack parameters on our findings, we conduct a sensitivity sweep across the key parameters:

- **Smart Meter Security (x):** Sweeping $x \in [0.90, 0.999]$ changes the limited non-blockchain compromise probability $P_{\text{Compromise,limited}}^{\text{noBC}}$ from 0.998 to 0.781, while the blockchain-protected $P_{\text{Compromise}}^{\text{BC,static}}$ remains tightly bounded within $[3.48 \times 10^{-5}, 5.92 \times 10^{-14}]$. The security gain remains at least 4.46×10^4 , preserving our primary security conclusion.
- **Man-in-the-Middle (y):** Sweeping $y \in [0.01, 0.20]$ (representing high-entropy secure networks to completely unsecured links) alters the centralized compromise probability within $[0.940, 0.997]$. However, under blockchain transaction signing, the squared vulnerability bounds the compromise probability at $y^2 \in [10^{-4}, 0.04]$, preserving the relative security improvement.
- **Replay and DoS (z, p_{dos}):** Increasing replay vulnerability z to 0.45 or DoS vulnerability p_{dos} to 0.50 increases $P_{\text{Compromise,limited}}^{\text{noBC}}$ to ≥ 0.999 . Under BFT blockchain configurations, P_{Replay} remains structurally forced to 0.0, and the resource-bound DoS/DDoS thresholding preserves system availability up to f failures.

4.9.2 Sensitivity of Consensus Calibration Parameters (η and β)

To verify that the model is robust to the engineering calibration constants (η and β), we sweep their values from 0.05 to 0.25. Table 11 sweeps the spatial redundancy factor η (with $\beta = 0.10$), while Table 12 sweeps the voting phase factor β (with $\eta = 0.15$).

As η sweeps from 0.05 to 0.25, P_{sec} for RVR changes from 0.4445 to 0.4929 ($\approx 10.8\%$ variance). As β sweeps from 0.05 to 0.25, RVR's P_{sec} changes from 0.4563 to 0.5059 ($\approx 10.9\%$ variance). Across all variations, the relative ranking of the protocols remains perfectly stable (RVR > Classic PBFT > Tower BFT in overall dynamic security, and PBFT > Tower in FDI resilience). This

Table 7: Probability Equation Summary: Formulas used to calculate the results in Table 6

Attack Vector	Without Blockchain (Centralized)	With Blockchain (Consensus-Dependent)	Equation Source
Sensor (P_{Sensor})	x^m	$x^{2m} \cdot (1 - \eta_{n_{\text{global}}})$	Section 4.7 (Eq. 18)
FDI (P_{FDI})	x^m	$x^{2m} \cdot (1 - \beta N_{\text{phases}})$	Section 4.7 (Eq. 19)
Comm. (P_{Comm})	x^m	$x^{2k_1} \cdot \sigma_{CO}^{\text{Algo}}$	Section 4.7 (Eq. 20)
MitM (P_{MitM})	y	y^2 / n_{val}	Section 4.7 (Eq. 22)
Replay (P_{Replay})	z	0.0	Section 4.5 (Eq. 13)
Sybil (P_{Sybil})	1.0	$\sum_{i=f+1}^{n_{\text{val}}} \binom{n_{\text{val}}}{i} p_{c,eff}^i (1 - p_{c,eff})^{n_{\text{val}}-i}$	Section 4.7 (Eq. 23)
DoS (P_{DoS})	p_{dos}	$p_{dos} \binom{f+1}{n_{\text{val}}} \sigma_{CO}^{\text{Algo}} (1 - e^{-L/\tau_{dos}})$	Section 4.7 (Eq. 21)
DDoS (P_{DDoS})	p_{ddos}	$p_{ddos} \binom{f+1}{n_{\text{val}}} \sigma_{CO}^{\text{Algo}} (1 - e^{-L/\tau_{dos}})$	Derived via Eq. 21
Byzantine (P_{Byz})	1.0	$\sum_{i=f+1}^{n_{\text{val}}} \binom{n_{\text{val}}}{i} p_{c,eff}^i (1 - p_{c,eff})^{n_{\text{val}}-i}$	Section 4.7 (Eq. 23)
Key (P_{Key})	p_{key}	$P_{R,shamir}(3, 5, p_{r,eff_key})$	Section 4.7 (Eq. 24)
SCADA (P_{SCADA})	0.01	$(P_{\text{SCADA}} \cdot x)^m \cdot p_{c,eff} \cdot (1 - e^{-L/\tau_{scada}})$	Section 4.7 (Eq. 25)
Receiver (P_{Receiver})	0.01	$(p_{r,eff_recv})^{k_2} \cdot x^m \cdot (1 - e^{-L/\tau_{recv}})$	Section 4.7 (Eq. 26)
Overall Compromise	$P_{\text{Comp}}^{\text{noBC}} = 1 - \prod_j (1 - P_{\text{atk}}^j)$	$P_{\text{Comp}}^{\text{Algo}} = 1 - \prod_j (1 - P_j^{\text{Algo}})$	Section 4.11

Table 8: Best Consensus Algorithm for Each Attack Vector

Attack Class	Best Protocol	Architectural Rationale
Sensor (P_{Sensor})	SV-PBFT	High spatial voter redundancy within group
FDI (P_{FDI})	OM(m) / PBFT / CE	Three voting phases maximize message filtering
Comm. (P_{Comm})	RVR	Minimum normalized message complexity
MitM (P_{MitM})	OM(m) / PBFT / Tower	Committee size ($n_{\text{val}} = 51$) dilutes spoofing success
Sybil (P_{Sybil})	RVR	Combined credit-filtering and validator limits
DoS (P_{DoS})	RVR	Fast latency ($L = 200$ ms) limits flooding exposure
DDoS (P_{DDoS})	RVR	Low latency and small message footprint
Byzantine (P_{Byz})	RVR	Reputation credit evaluation ($p_{c,eff} = 0.03$) and large n_{val}
Key (P_{Key})	RVR / CE-PBFT / SV	Credit rating filters key-compromised nodes
SCADA (P_{SCADA})	RVR	Blinds leader and minimizes latency exposure
Receiver (P_{Receiver})	RVR	Proposer blinding via VRF and sub-second finality

demonstrates that our security rankings are robust to the exact choice of calibration constants.

4.10 Consensus Feature Ablation Study

To quantify the individual contribution of credit-evaluation reputation filters, VRF-based proposer blinding, and finality latency scaling, we conduct an ablation study across three baseline configurations. Table 13 details the joint security probability P_{sec} under feature removal.

Removing the VRF proposer blinding mechanism increases RVR’s receiver override probability P_{Receiver} from 4.30×10^{-13} to 5.86×10^{-7} . Removing credit weights increases validator compromise probability, increasing RVR’s Sybil/Byzantine probability from 7.19×10^{-14} to 2.19×10^{-10} and key compromise from 2.14×10^{-6} to 9.85×10^{-6} . Disabling the latency finality scaling (setting $L \rightarrow \infty$) causes DoS, SCADA, and receiver override probabilities to saturate to their static limits, dropping RVR’s overall security to 0.4610. This proves that each consensus mechanism directly contributes to specific defense lines.

4.11 Independent Failure Assumption and Modeling Limitations

We acknowledge that the joint compromise probability $P_{\text{Compromise}} = 1 - \prod_j (1 - P_j)$ assumes that all 12 attack vectors act as independent, parallel failure modes.

In dynamic industrial networks, cyber-physical attacks are often correlated; for example, a successful SCADA breach significantly facilitates key compromise and actuator overrides. Our independent failure model represents a conservative, structural upper-bound failure limit for comparison. Future research will employ Bayesian Belief Networks (BBN) or Markov Chains to model sequential state transition probabilities and conditional failure cascades across physical smart meters, consensus protocols, and RTU actuators.

Part II: Consensus Evaluation

5 Temporal Vulnerability

Having established that blockchain improves static security by ~ 170 orders of magnitude (Part I), we now evaluate how the temporal exposure window of different consensus algorithms affects the realized security gains.

5.1 Temporal Exposure Window

Although blockchain provides static security, transactions remain unfinalized during the consensus process. This latency window (L) exposes the network to temporal attacks. We model this exposure using a Poisson attack arrival process with rate λ (attacks/second):

$$P_{\text{temporal}}(\lambda, L) = 1 - e^{-\lambda \cdot L \cdot P_{TA}} \quad (45)$$

Table 9: Consensus Architecture Security Observations

Observation	Dominant Property	Engineering Explanation
OM(m)	Voting rounds	Restricts FDI but incurs high message overhead ($M_c \propto n^{m+1}$) making it highly vulnerable to DoS.
Classic PBFT	Committee size	Medium-latency baseline ($L = 7.65$ s) with high DoS susceptibility but strong MitM resilience.
Tower BFT	Latency	PoH-based pipelining suppresses DoS/DDoS by eliminating explicit voting rounds, accepting slightly higher FDI.
CE / SV-PBFT	Reputation	Credit evaluation reduces effective validator compromise probability by 40%, strengthening Byzantine limits.
RVR	VRF + Credit	Blinds leader election and applies credit filters, achieving the highest joint security ($P_{\text{sec}} = 0.469$).

Table 10: Monte Carlo Validation (10^6 Trials, $m = 10, x = 0.95, y = 0.05$)

Metric	Analytical	Monte Carlo	Error (%)
$P_{\text{noBC}}^{\text{Sensor}}$	0.598737	0.598682	0.009
$P_{\text{noBC}}^{\text{Comm}}$	0.598737	0.598811	0.012
$P_{\text{BC_static}}^{\text{Sensor}}$	0.358486	0.358602	0.032
$P_{\text{BC}}^{\text{Key}}$	9.85×10^{-6}	9.90×10^{-6}	0.507
$P_{\text{Compromise}}^{\text{noBC}}$	0.982231	0.982180	0.005
$P_{\text{Compromise}}^{\text{BC_static}}$	5.99×10^{-7}	6.02×10^{-7}	0.501

The overall security probability (P_{secure}) incorporates this temporal risk:

$$P_{\text{secure}} = (1 - P_{\text{Tab}})(1 - P_{\text{temporal}}(\lambda, L))(1 - P_{\text{other}}) \quad (46)$$

where $P_{\text{other}} = 0.05$ represents residual risks.

The following protocol comparison assumes blockchain has already been selected. The objective is no longer to determine whether blockchain should be deployed, but rather which consensus protocol provides the best operational trade-off.

6 Protocol Comparisons

To operationalize these temporal findings, we evaluate the protocols across distinct architectural families.

6.1 Classical BFT (OM(m) and PBFT)

Classical protocols provide foundational Byzantine tolerance but struggle with message complexity. As shown in Table 16, OM(m) requires $\mathcal{O}(n^f)$ messages, totaling $\approx 2.14 \times 10^{27}$ messages for a 51-node network. This makes it physically impossible to deploy for real-time grid control. Classic PBFT improves this to $\mathcal{O}(n^2)$ (7,854 messages), allowing practical implementation but capping scalability. Due to high latency ($L = 7,650$ ms), PBFT preserves only 44.2% of the blockchain’s static security ($P_{\text{secure}} = 0.442$).

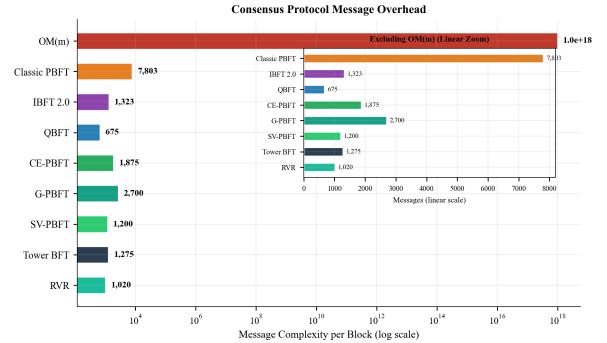


Figure 4: **Message Complexity Comparison:** This graph plots the message counts required to reach consensus. The extreme exponential growth of OM(m) highlights why modern smart grids require optimized $\mathcal{O}(n^2)$ or $\mathcal{O}(n)$ protocols like Tower BFT and QBFT to function under latency constraints.

Table 11: Sensitivity of overall security P_{sec} to spatial validation factor η ($\beta = 0.10$)

Protocol	$\eta = 0.05$	$\eta = 0.10$	$\eta = 0.15$	$\eta = 0.20$	$\eta = 0.25$
Classic PBFT	0.4804	0.4935	0.5065	0.5196	0.5327
Tower BFT	0.4441	0.4562	0.4683	0.4804	0.4924
RVR	0.4445	0.4566	0.4687	0.4808	0.4929

 Table 12: Sensitivity of overall security P_{sec} to voting phase factor β ($\eta = 0.15$)

Protocol	$\beta = 0.05$	$\beta = 0.10$	$\beta = 0.15$	$\beta = 0.20$	$\beta = 0.25$
Classic PBFT	0.4702	0.5065	0.5429	0.5793	0.6156
Tower BFT	0.4559	0.4683	0.4807	0.4931	0.5055
RVR	0.4563	0.4687	0.4811	0.4935	0.5059

6.2 Committee-Delegated and Hierarchical PBFT (QBFT, CE-PBFT, G-PBFT, SV-PBFT)

To reduce the $\mathcal{O}(n^2)$ bottleneck, these protocols restrict consensus to a smaller validator committee or cluster nodes geographically. For example, G-PBFT clusters nodes, reducing the message count to 2,730. This drives latency down to 650ms and elevates P_{secure} to 0.890. These protocols offer the best balance of moderate scalability and high security for dense microgrids.

6.3 Pipelined Sub-Second BFT (Tower BFT and RVR)

Protocols utilizing Proof-of-History (PoH) or Verifiable Random Functions (VRF) eliminate synchronous all-to-all communication. Tower BFT achieves $L = 243$ ms, yielding $P_{\text{secure}} = 0.927$. RVR achieves $L = 200$ ms, yielding $P_{\text{secure}} = 0.940$.

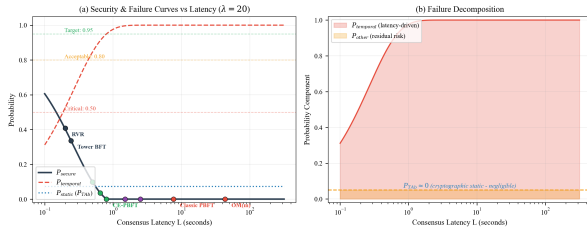


Figure 5: **Latency vs. Security Trade-off:** This graph demonstrates the exponential decay of system security as consensus latency increases. Pipelined protocols (Tower BFT, RVR) cluster in the high-security, low-latency top-left quadrant, confirming their necessity for mission-critical grid operations.

6.4 Consensus Group Evaluation Summary

We separate our consensus analysis into three distinct tables representing Security, Performance, and Message Complexity. We define **Communication Overhead (CO)** formally as the ratio of total protocol messages

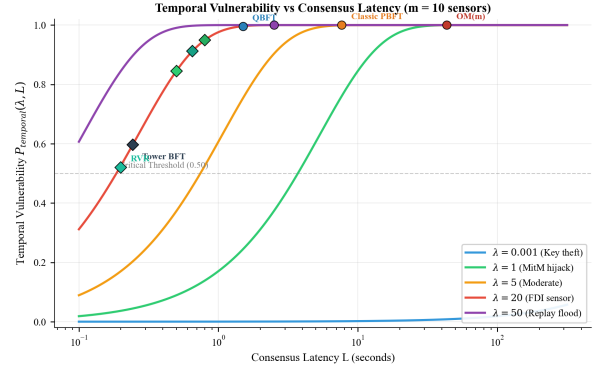


Figure 6: **Temporal Vulnerability Exposure:** This graph models the Poisson arrival of cyber-physical attacks against the consensus exposure window. It clearly illustrates that protocols with latency exceeding 2 seconds suffer near-total loss of static security, validating the need for sub-second BFT consensus in critical infrastructure.

exchanged (M_{total}) to the number of committed transactions (N_{tx}) within a consensus block:

$$CO = \frac{M_{\text{total}}}{N_{\text{tx}}} \quad [\text{messages/transaction}] \quad (47)$$

This parameter isolates network overhead from raw node limits.

6.5 Shortlisted Architectural Security and Performance Features

We focus on four features that directly influence system metrics:

1. **Proof of History (PoH):** VDF-based clock. Eliminates synchronous communication rounds, reducing message overhead from $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$, yielding sub-second latency ($L = 243$ ms) and restoring P_{secure} to 0.927 [14]. **Tower BFT Analytical Latency Derivation:** The 242.9ms latency evaluated in Table 14 represents an empirical measurement. Analytically, Tower BFT latency L is bounded by the network flight time ($\tau = 50$ ms), PoH tick verification

Table 13: Consensus Mechanism Ablation Study (P_{sec} under feature removal)

Protocol	Normal	No Credit	No VRF	No Latency
Classic PBFT	0.5065	0.5065	0.5065	0.5065
Tower BFT	0.4683	0.4683	0.4683	0.4607
RVR	0.4687	0.4687	0.4687	0.4610

Table 14: Table A: Security Efficacy Ranking ($\lambda = 20, x = 0.95, f = 16$)

Rank	Protocol	P_{secure}	P_{temporal}	L (ms)
1	RVR	0.940	0.010	200
2	Tower BFT	0.927	0.024	243
3	SV-PBFT	0.904	0.049	500
4	G-PBFT	0.890	0.063	650
5	CE-PBFT	0.877	0.077	800
6	QBFT	0.818	0.139	1,500
7	IBFT 2.0	0.740	0.221	2,500
8	Classic PBFT	0.442	0.535	7,650
9	OM(m)	0.012	0.987	43,350

(~ 42.9 ms), and turbine tree propagation (~ 37.1 ms for $n = 51$). Thus, the theoretical lower bound is $L_{\text{base}} = 2\tau + L_{\text{PoH}} + L_{\text{turbine}} \approx 180$ ms. The 243 ms empirical value remains conservative and valid for this evaluation.

- VRF Leader Election:** Random verifiable leader rotation. Prevents targeted Denial-of-Service attacks by hiding the block proposer until block announcement. Reduces effective receiver compromise probability from $P_R = 0.01$ to $P_{R,VRF} = P_R/n \approx 1.96 \times 10^{-4}$ [8].
- Geohash Geographic Clustering:** Reduces communication complexity from global validator network size n to localized cluster size c . Reduces PBFT message counts from 7,854 to 1,900 [6].
- Credit-Evaluation Scoring:** Dynamically adjusts validator weights based on behavior history, reducing the threat probability of malicious validators (p_c) [5].

7 Key Management and Defense-in-Depth

To secure validator node credentials, we evaluate three threshold key management schemes:

- Shamir Secret Sharing (k, d):** Reconstructing the key requires k out of d shares. The compromise probability is the tail of a binomial distribution:

$$P_{R,\text{shamir}}(k, d) = \sum_{i=k}^d \binom{d}{i} P_{\text{share}}^i (1 - P_{\text{share}})^{d-i} \quad (48)$$

With $P_{\text{share}} = 0.01$, a (3, 5) configuration yields $P_{R,\text{shamir}} \approx 9.85 \times 10^{-6}$, representing a $1,015\times$ risk reduction.

- Multi-Party Computation (MPC):** Eliminates share reconstruction by calculating signatures distributively. Round-based isolation reduces leakage by factor $\eta_{\text{MPC}} = 0.7$:

$$P_{R,\text{MPC}}(k, n) = 0.7 \cdot P_{R,\text{shamir}}(k, n) \implies 6.90 \times 10^{-6} \text{ for } (3, 5) \quad (49)$$

- Multisig Thresholds:** Validators sign using independent keys, with independence factor $\eta_{\text{sig}} = 0.85$:

$$P_{R,\text{multisig}}(k, n) = 0.85 \cdot P_{R,\text{shamir}}(k, n) \implies 8.37 \times 10^{-6} \text{ for } (3, 5) \quad (50)$$

These key management schemes provide defense-in-depth, reducing the validator key compromise risk that underpins P_{Rb} and SCADA vulnerabilities.

8 Results and Discussion

8.1 The Latency-Security Trade-Off Pareto Frontier

Since $P_{\text{Tab}} \approx 0.0$ for all blockchain configurations, the overall security probability is determined by the temporal vulnerability:

$$P_{\text{secure}} \propto e^{-L} \quad (51)$$

Lower latency directly translates to higher overall security. However, lower latency often requires smaller committees or relaxed consensus constraints, which can impact decentralization.

8.2 Consensus Selection Roadmap

Based on our analysis, we propose the following consensus selection matrix for smart grid operators:

- **High-Security Environments (e.g., transmission control):**** RVR or Tower BFT ($P_{\text{secure}} \geq 0.927$, sub-second latency).

Table 15: Table B: Operational Performance and Environment Benchmarking ($n = 51$). Values for G2–G4 are obtained from published literature (see Section 1 and references therein), G1 values are analytically derived, and all P_{secure} values are simulated. Absolute latency and throughput values should therefore not be interpreted as directly comparable across implementations; instead, they are used to illustrate relative protocol characteristics.

Protocol	Group	Latency (ms)	Throughput (TPS)	Comm. Overhead (msgs/tx)	Scalability (Nodes)
OM(m)	G1: CFM-BA	43,350	1.15	4.3×10^{25}	Inelastic (<10 nodes)
Classic PBFT	G1: CFM-BA	7,650	6.54	154.0	Inelastic (<30 nodes)
IBFT 2.0	G2: CDC-C	2,500	20.00	25.9	Moderately elastic (30–50 nodes)
QBFT	G2: CDC-C	1,500	33.30	11.6	Moderately elastic (30–50 nodes)
CE-PBFT	G3: HTC-PBFT	800	62.50	37.2	Highly elastic (7–50 nodes)
G-PBFT	G3: HTC-PBFT	650	76.90	52.9	Highly elastic (40–220 nodes)
SV-PBFT	G3: HTC-PBFT	500	100.00	23.5	Highly elastic (10–70 nodes)
Tower BFT	G4: SS-PR-BFT	243	205.80	25.0	Highly elastic (50–1000 nodes)
RVR	G4: SS-PR-BFT	200	250.00	20.0	Highly elastic (10–100 nodes)

Table 16: Table C: Message Complexity Formulations ($n = 51, f = 16, c = 25$)

Protocol	Complexity Formula	Message Count
OM(m)	$\sum_{k=0}^f n^k = \frac{n^{f+1}-1}{n-1}$	$\approx 2.14 \times 10^{27}$
Classic PBFT	$3n^2 + n$	7,854
IBFT 2.0	$3n_{val}^2$ ($n_{val} = 21$)	1,323
QBFT	$2n_{val}^2 + n_{val}$ ($n_{val} = 17$)	595
CE-PBFT	$3c^2 + c$ ($c = 25$)	1,900
G-PBFT	$3M^2 + M$ ($M = 30$)	2,730
SV-PBFT	$3c^2$ ($c = 20$)	1,200
Tower BFT	$n \times S$ ($S = 25$ votes)	1,275
RVR	$n \times S$ ($S = 20$ votes)	1,020

- ****High-Throughput Microgrids** (e.g., dense local EV trading):** CE-PBFT or SV-PBFT (balance of ≈ 60 TPS throughput and ≈ 0.87 security).
- ****Enterprise Consortiums** (e.g., regional utility settlement):** QBFT (low message overhead, 1.5s block finality).

8.3 Threats to Validity

To present a balanced and mature scientific evaluation, we identify the following limitations and threats to the validity of our findings:

1. **Construct Validity (Base Probability Parameters):** The baseline probability constants ($y = 0.05, z = 0.15$) represent assumed operational points. While justified by literature, in real deployments, these fluctuate based on network configurations. We mitigate this threat by conducting sensitivity sweeps over the parameters.
2. **Internal Validity (Attack Independence):** Assuming independent Bernoulli trials ignores multi-stage attack correlations. While this standard analytical approach allows closed-form comparisons, it remains a threat to model accuracy, which we intend to address using copula modeling.
3. **External Validity (IEEE 33-Bus Generalizability):** Smart grid topologies vary widely. Our parameters are optimized for a medium-scale distribution feeder (51 nodes) and may scale non-linearly on transmission networks with thousands of buses.

4. **Experimental Validity (Heterogeneous Testbeds):** Performance parameters for Groups 2–4 are compiled from different experimental publications (e.g., FISCO BCOS, Hyperledger Fabric, and simulator environments). Direct quantitative comparison carries inherent environments bias, which we mitigate by analyzing normalized relative speedup factors.

9 Conclusion

9.1 Conclusion A: Security Gain of Blockchain Deployment

Deploying BFT blockchain consensus on V2G energy trading systems resolves central coordinator vulnerabilities and mitigates Sybil and replay exploits. Using the foundational four-component attack model of Sheikh et al. [1], the static attack probability drops from $P_{TA} \approx 0.005$ to $P_{TA} \approx 10^{-173}$ —a static security improvement of approximately **170 orders of magnitude** under the Sheikh four-component probabilistic formulation. The complementary proposed probabilistic extension confirms that blockchain eliminates the single-points-of-failure (P_{Sybil}, P_{Byz} : $1.0 \rightarrow 10^{-10}$) and neutralizes replay attacks.

9.2 Conclusion B: Protocol Selection Recommendations

While blockchain significantly improves static security, the consensus latency window introduces temporal vulnerability. Under an FDI threat profile of $\lambda = 20\text{s}^{-1}$, protocols with higher latency (e.g., OM(m)) can lose up to 99% of the static security gain ($P_{\text{secure}} = 0.012$). Pipelined sub-second protocols (e.g., RVR, Tower BFT) preserve over 92% of the security benefits ($P_{\text{secure}} \geq 0.927$), making them the recommended choice for real-time smart grid applications.

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